Beyond Threshold Adaptation: Error Trade-offs in Intrusion Detection with Sparse Delayed Labels

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Abstract

How much of an adaptive intrusion detector's apparent improvement comes from moving its decision threshold, and where does the remaining benefit occur? We examine this question using paired chronological replays with sparse delayed labels. A four-trace experiment separates a frozen or refitted additive spline head from a historical or recent threshold. On the original TON_IoT evaluation, threshold adaptation alone raises recall from 21.1% to 89.1%; adding head adaptation mainly reduces false positives. However, 96.7% of that reduction occurs on feature vectors observed in earlier intervals, where fewer false alarms are accompanied by more missed attacks. Only three of five seeds meet the pre-specified joint-error criterion. A post-review extension adds an independently updated threshold, a threshold using the full acquired-label budget, intercept-only and prior-centred controls, logistic regression, an adaptive random forest, delay and budget sensitivity, and a separate chronological UNSW-NB15 evaluation. Thresholds for five error-cost ratios are selected exclusively from available past labels. The spline head changes the FN/FP compromise in dataset-dependent directions, whereas the forest retains substantial benefit over its own frozen control. The contribution is an auditable decomposition of decisions, ranking, label availability, and representation overlap, rather than a new model architecture or a universal advantage of adaptation.

Keywords: network intrusion detection; concept drift; delayed labels; decision thresholds; error trade-offs; additive splines; reproducible evaluation.

1 Introduction

Updating an intrusion detector can change two different objects: the score used to rank observations and the decision threshold used to raise alarms. A detector may recover missed attacks by changing only its threshold. Conversely, an updated score can rank attacks more accurately while producing poor decisions at an inherited threshold. A useful evaluation must establish which intervention changes false positives (FP) and false negatives (FN), and which observations receive the benefit.

These distinctions matter under limited supervision. Concept-drift methods respond to changes in the data-generating process [1], while prequential evaluation requires prediction before learning from an observation [2]. Security labels can be expensive and delayed; their availability constrains both refitting and threshold selection. Operational usefulness also depends on attack prevalence and alarm workload, not just a balanced benchmark score [3, 4].

We ask: *how much of the apparent gain over a historical detector is reproduced by threshold adaptation alone, and how does the incremental effect of learning differ across operating costs and previously observed versus unobserved feature vectors?* Our initial case study uses a compact additive spline representation with nine refitted head parameters. It provides an explicit intervention rather than an architectural novelty. A subsequent extension includes logistic regression and adaptive random forests to test how the observed pattern depends on the learner.

The study has two evidence stages. The original TON_IoT campaign follows a protocol fixed after development and retains its unsuccessful four-of-five joint FN/FP acceptance criterion. The post-review extension is specified before its new runs, but its TON results are exploratory because the original evaluation interval was already examined. It adds stronger threshold-only controls and a separate chronological UNSW-NB15 capture. Across both stages, we report effective label use, threshold eligibility, absolute errors, overlap strata, ranking, and causally selected operating points. This makes the contribution a reusable evaluation procedure and an empirical account of conditional trade-offs. The forest comparator supplies a useful counterexample to a threshold-only explanation: its learning benefit remains substantial against its own frozen baseline.

2 Related work and scope

2.1 Adaptive security models with limited and delayed labels

Sparse supervision and delayed feedback are established problems in stream learning. Active learning under drift studies how to allocate a limited labeling budget [5]; adaptive random forests combine online bagging, feature randomization, and drift-triggered replacement [6]. Delayed-label evaluation explicitly separates observation, prediction, and learning times [7]. We retain uniform sampling to hold label acquisition fixed while comparing model and threshold updates.

INSOMNIA directly studies network intrusion detection under drift, using active learning and label estimation to reduce update latency and labeling overhead [8]. In Android security, continuous learning combines contrastive representations with selective labeling [9]. These methods address how to acquire information or adapt a model. Our complementary question concerns attribution: how much of a reported decision gain survives a stronger threshold-only baseline using the same acquired labels?

Transcend detects aging malware classifiers through comparisons with their training distributions [10]. CADE identifies and explains individual drifting samples using a learned contrastive representation, including a network-intrusion case study [11]. These are drift-detection and explanation mechanisms; our paired update controllers examine the downstream error trade-off once delayed labels become available.

Temporal consistency also extends beyond a chronological train/test boundary. TESSERACT formalizes spatial and temporal biases in security evaluation [12]; broader security-ML analyses emphasize experimental assumptions and deployment mismatches [4, 13]. Here label timestamps are explicit within the test interval, and exact vector overlap is reported separately from row identity. Dataset audits such as the CICIDS2017 case study show why preprocessing and label provenance need independent checks [14].

2.2 Threshold selection, prior shift, and ranking

Probability calibration estimates agreement between predicted probabilities and observed frequencies [15]. In this paper, CAL denotes a memory used to select a binary decision threshold. It does not imply calibrated posterior probabilities. AUROC measures ranking and is unchanged by a threshold-only intervention on fixed scores.

Changes in class proportions can also change a classifier's operating point. Prior-probability correction is well established [16], and label-shift estimation assumes that class-conditional feature distributions remain stable while class priors change [17]. Class-balanced refitting makes a score shift plausible in our protocol. However, changing attack families, feature distributions, head slopes, and regularization prevent attributing the observed shift to pure label shift. We therefore treat a refitted head with its historical threshold as a diagnostic control and test a separate intercept-only correction.

2.3 Additive models and intrusion datasets

An additive logistic predictor with smooth univariate terms belongs to the generalized additive model tradition [18]. Neural additive models learn such terms with neural subnetworks [19]; KANs use learnable univariate edge functions [20]. Our fixed B-spline representation can be described as KAN-style, but the online intervention is a linear head refit over eight fixed feature contributions. Results from convolutional KAN intrusion classifiers [21] therefore provide architectural context rather than a matched performance baseline.

TON_IoT provides heterogeneous IoT/IIoT network data [22, 23]. UNSW-NB15 supplies a separate laboratory capture with different normal traffic and attack composition [24, 25]. We use compact numerical feature sets and complete-flow availability in both. The datasets provide two empirical settings; their controlled captures and feature contracts delimit the scope of generalization.

3 Materials and methods

3.1 Dataset, feature availability, and chronological partitions

The source network archive contains 22,339,021 rows. The preprocessing manifest records 869 invalid rows and 22,338,152 retained rows. The retained input vector contains eight numerical fields in this order: `duration`, `src_bytes`, `dst_bytes`, `missed_bytes`, `src_pkts`, `src_ip_bytes`, `dst_pkts`, and `dst_ip_bytes`. Binary labels distinguish normal traffic from attacks. The eight-feature contract was inherited from the project's compact-detector experiments: it uses flow-level numerical aggregates without IP identifiers, ports, application strings, or categorical encodings. Holding it fixed makes the adaptation intervention comparable, but is not a feature-optimality claim. The chronological evaluation stream retains its observed prevalence.

Records are ordered by their complete-feature availability time,

$$t_{\text{ready}} = t_{\text{start}} + \text{duration}. \tag{1}$$

This is an offline flow-completion replay. It does not assume that duration or byte totals are known at the first packet. The retained data cover ten UTC calendar days in April 2019, with no retained interval from 5 to 22 April. Table 1 defines the four chronological partitions. A estimates the initial transformation and model; B selects the initial threshold; C supports development; and D is reserved for the final predictive campaign.

Attack composition changes across these intervals. A contains scanning attacks; B contains scanning and denial of service. C contains distributed denial of service, denial of service, injection, and password attacks. D contains backdoor, man-in-the-middle, password, ransomware, and cross-site scripting records. These changes provide temporal heterogeneity, but the experiment does not identify a causal type of drift. Backdoor, man-in-the-middle, ransomware, and cross-site scripting are absent from A/B/C; password also occurs in C. The extension reports per-family attack recall.

Before the final campaign, D had been read for structural checks and feature-vector overlap analysis. Its predictive evaluation was run only after the protocol and code were fixed. Accordingly,

Table 1: Chronological data partitions. Counts refer to retained rows; normal and attack counts sum to each partition total. The separate overlap strata are described in Section 3.5.

Partition	UTC dates, April 2019	Total	Normal	Attack
A: initial fit	2–4, 23	3,956,308	223,199	3,733,109
B: initial threshold	24	5,681,198	89,323	5,591,875
C: development	25–26	9,150,673	173,316	8,977,357
D: final evaluation	27–29	3,549,973	309,673	3,240,300

D is a later evaluation interval, not a completely unseen resource in the broader research process. The five initial models and their B thresholds are prepared before the final runner loads D. C is used in that runner to construct the prior-vector inventory; the C adaptive state is not carried into D.

3.2 Initial model and restricted adaptation

Each raw input is transformed by $\log(1 + \max(x, 0))$ and standardized using means and standard deviations estimated from A. A zero standard deviation is replaced by one. The transformation remains fixed. Each transformed feature is then mapped to its range observed in A and clipped to $[0, 1 - 10^{-9}]$ for spline evaluation. Values outside the observed range therefore saturate at a boundary rather than extrapolating the spline.

For input j, the model contains eight cubic B-spline basis functions B_{jr} with an open knot vector based on six equally spaced positions between zero and one. The initial function is

$$f_j(x_j) = \sum_{r=1}^{8} \theta_{jr} B_{jr}(u_j(x_j)), \qquad s(x; g, b) = b + \sum_{j=1}^{8} g_j f_j(x_j). \tag{2}$$

The 64 spline coefficients and initial intercept are fitted by logistic regression on a uniform, seeded 400,000-row sample of A. The implementation uses scikit-learn with `C=1.0` and `max_iter=1000` [26]. The initial gains are $g_j = 1$. Subsequent adaptation leaves the transformation, knots, and all θ_{jr} fixed, and refits only the eight gains and intercept on the current fitting memory. Each refit solves a new logistic-regression problem with L2 shrinkage of the eight weights toward zero and an unpenalized intercept. It neither starts from nor penalizes distance to the previous head. The fixed spline functions retain information from A, while the head is newly fitted to the current memory. The extension explicitly tests a penalty centred at the previous head.

The empirical object is a logistic generalized additive model with an adaptive linear combination of fixed spline functions. Head refitting can reweight these functions, but cannot learn a new spline shape or an interaction between inputs.

3.3 Threshold objective and four paired traces

The implementation predicts attack when $s(x) > \tau$ and normal otherwise. For labeled set $\mathcal{L}$, the threshold objective is

$$\tau^*(\mathcal{L}, s) \in \arg\max_{\tau \in \mathcal{T}} \frac{1}{2} \left[\frac{\mathrm{TP}(\tau)}{P} + \frac{\mathrm{TN}(\tau)}{N} \right], \tag{3}$$

where P and N are the attack and normal counts in $\mathcal{L}$. Candidates comprise the distinct observed scores and a floating-point value strictly below the minimum; together with the maximum observed score, this covers the all-attack and all-normal decisions. Scores and objective values use float64. Maximizers are selected by exact equality, without a tolerance, and ties are resolved by the lower median of the sorted maximizing candidates, at zero-based index $\lfloor (m-1)/2 \rfloor$. This is a deterministic convention, not a demonstrated threshold-stability property.

Table 2: Four paired traces. B denotes the initial fixed threshold; CAL64 denotes recent threshold selection. The primary contrast is the last row minus the second row.

Trace	Score function	Decision threshold
Frozen / B	Initial A model	Fixed, selected on B
Frozen / CAL64	Initial A model	Selected on recent CAL64 labels
FIT192 / B	Head refitted on FIT192	Fixed, selected on B
FIT192 / CAL64	Same refitted head as FIT192 / B	Selected on recent CAL64 labels

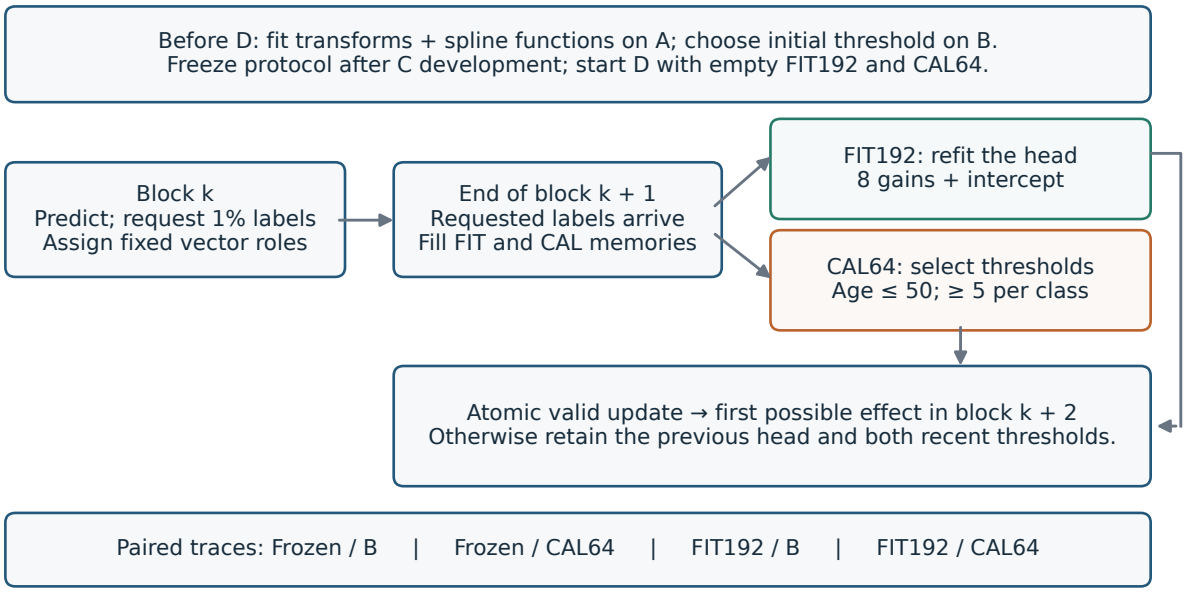


Figure 1: Causal protocol. A label requested in block k cannot affect predictions before block $k+2$. Fitting and threshold selection use separate memories; all state changes are committed together only after the eligibility checks succeed.

The B threshold is selected separately for each initial model using all of B, then kept fixed in the corresponding B traces. Table 2 defines the four traces. Both frozen traces use identical initial scores; both FIT192 traces use an identical adapted score trajectory. Consequently, a threshold-only contrast cannot change AUROC. All traces share the same label requests and update eligibility decisions within a seed.

3.4 Label budget, delay, and memory separation

D is divided into 354 blocks of 10,000 records and a final block of 9,973 records, indexed from 0 to 354. Predictions for a block are emitted before any labels from that block enter the controller. A uniform sample without replacement requests $\lfloor 0.01 n_k \rfloor$ labels from block k: 100 from a full block and 99 from the last block. Requested labels arrive at the end of block $k+1$ and can first influence predictions in block $k+2$ (Figure 1). Thus 35,499 labels are requested, of which the final 99 remain pending when the observed stream ends. Full labels are available only to the offline evaluation code; the online update receives the delayed requested subset.

Both memories start empty at the beginning of D. FIT192 holds at most 192 labeled examples, with target quotas of 96 normal and 96 attack examples. CAL64 holds at most 64 examples, with target quotas of 32 per class. Either class can borrow unused capacity. On overflow, the oldest example from a class above quota is removed. The quotas alter the stored fitting and

threshold-selection samples, not the evaluation stream or the requested-label budget.

A fixed salted SHA-256 rule assigns each raw eight-feature vector permanently to one role: digest modulo four equal to zero selects CAL, and other values select FIT. This prevents an identical vector from appearing in both fitting and calibration memories, even if it recurs later in the stream. It does not deduplicate either memory internally. Role serialization uses canonical float64 bytes; exact overlap between chronological partitions is measured separately using the stored float32 vectors. Appendix A specifies these encodings.

FIT192 requires both binary classes. CAL64 contributes only entries observed at most 50 blocks earlier and requires at least five normal and five attack labels. FIT192 has no corresponding age cutoff. If these conditions hold, the implementation fits a candidate head, scores the eligible CAL64 examples with both the frozen and candidate heads, and computes two thresholds using Equation 3. A valid candidate head and both thresholds replace the prior state atomically for the next block. A failed eligibility or numerical check leaves all prior state unchanged and records the reason.

This common schedule isolates the effect of enabling head fitting within the fixed controller, while keeping threshold data and update eligibility matched. It does not estimate a benefit per additional label: FIT192/CAL64 can use FIT-role labels that Frozen/CAL64 leaves unused. The extension therefore adds independently eligible CAL64 and a frozen CAL256 control that can use every acquired label. The 256-example capacity also excludes the delayed-label queue, model state, feature transformations, Python object overhead, temporary solver arrays, and full input arrays used by the offline runner. It must not be interpreted as a total memory bound for an embedded implementation.

3.5 Overlap-aware reporting

The chronological partitions have disjoint row identities, but their eight-feature vectors overlap. An exact SHA-256 inventory of A/B/C marks each D vector as previously observed or unobserved. In D, 37.23% of rows and 58.03% of normal rows match a prior vector. The unobserved-vector mask retains 2,228,436 rows: 129,982 normal and 2,098,454 attack observations. The complementary seen stratum contains 179,691 normal and 1,141,846 attack observations.

This mask is applied only when aggregating predictions. It changes neither the prediction sequence nor the adaptive state. Requested labels from all of D can still enter the controller, and repetitions within D remain. The supplementary result therefore measures performance on vectors absent from A/B/C during the same adaptive replay. It is not a separately trained no-overlap experiment, an unseen-attack-family test, or a guarantee that a vector has never previously affected the state within D. A matching eight-feature vector is also not proof that two original network events are duplicates.

3.6 Metrics, paired seeds, and decision criterion

We report global false negatives (FN), false positives (FP), attack recall TP/P, false-positive rate FP/N, and balanced accuracy

$$\mathrm{BA} = \tfrac{1}{2}\left(\mathrm{recall} + 1 - \mathrm{FPR}\right). \tag{4}$$

These rates are computed from the confusion matrix aggregated over the selected scope. Ranking is summarized by the unweighted mean of AUROC over blocks containing both classes:

$$\overline{\mathrm{AUC}}_{\mathrm{block}} = \frac{1}{|\mathcal{V}|}\sum_{k\in\mathcal{V}} \mathrm{AUC}_k. \tag{5}$$

Undefined AUROCs remain missing. Valid denominators are 352 of 355 blocks for all D and 350 of 355 for the A/B/C-unseen subset. This statistic is not the AUROC obtained by pooling every

Table 3: Additional controls. Every row shares the acquisition budget and label delay. Retained examples exclude model state and the pending-label queue.

Controller	Learning information	Threshold information and eligibility
Frozen / independent CAL64	None	CAL-role labels; independent CAL gate
Frozen / CAL256	None	All queried labels; independent CAL gate
Offset / B or CAL64	FIT192; one fitted intercept offset	B or disjoint CAL64; matched head gate
Prior-centred / CAL64	FIT192; eight weights and intercept	Disjoint CAL64; matched head gate
LR / FIT192 / CAL64	FIT192; eight linear weights and intercept	Disjoint CAL64; matched head gate
ARF / CAL64	Every arriving queried label, once	CAL-role labels, also used for learning; independent CAL gate

score across the stream; it does not require scores from different adaptive states to be directly comparable.

Seeds 42–46 determine the initial A subsample and block-level label sampling. The original sampler initializes each block with seed+k; different seed/block pairs can therefore reuse the same relative query positions. The extension replaces this with a tuple-keyed SeedSequence while retaining common requests across methods. Each seed shares all observations and requests across its four traces. Values are means and sample standard deviations ($n-1$ denominator) over the five seeds. Differences are computed within seed before aggregation. These deviations describe stochastic variation conditional on one recorded stream; they are neither confidence intervals nor variation across five independent datasets.

The versioned protocol fixed the primary comparison as FIT192 / CAL64 minus Frozen / CAL64. Its decision criterion required $\Delta\text{FN} \leq 0$ and $\Delta\text{FP} \leq 0$, with at least one strict inequality, in at least four of five seeds. This is a pre-specified empirical acceptance rule, not a statistical significance test or an externally registered trial. The full-stream and supplementary outcomes are reported whether or not the criterion succeeds. We do not select a new operating threshold using final D outcomes.

3.7 Post-review extension: stronger controls and operating costs

The extension protocol was committed before its new predictive runs. It retains seeds 42–46, 10,000-row blocks, and the original memory rules. A new sampler uses `SeedSequence([seed, block, 20261101])`; all methods receive identical requested row IDs within each condition. A separate legacy-sampler replay provides a bridge to the earlier campaign. The original numbers remain reported as originally measured, rather than being overwritten by reruns.

Table 3 distinguishes the comparisons. CAL256 has quotas of 128 observations per class, accepts every acquired label, and uses the same 50-block age limit and five-per-class eligibility minimum. Independent CAL64 removes only the head-fitting eligibility condition. These controls separate equality of acquired-label budgets from equality of the labels actually used by an algorithm.

The offset control minimizes logistic loss for $s_0(x)+c$ on FIT192. With a freely selected threshold, $s_0(x)+c > \tau+c$ gives the same decision as $s_0(x) > \tau$; an intercept-only update cannot improve the ranking or enlarge the attainable set of decisions. Its fixed-B trace is informative about operating-point movement, while its CAL64 trace tests this invariance in the implementation.

For the prior-centred control, at each eligible update we minimize

$$\sum_{i\in\mathrm{FIT192}}\Big[\log(1+e^{w^T r_i+b})-y_i(w^T r_i+b)\Big]+\tfrac{1}{2}\|w-w_{\mathrm{previous}}\|_2^2. \tag{6}$$

The intercept is unpenalized. The penalty strength is fixed at one, matching the summed-loss scale of the original $C=1$ refit. This changes the optimization objective; merely warm-starting a zero-centred optimizer would not implement the same control.

Logistic regression uses the eight standardized log-transformed inputs directly. Its initial fit uses the same 400,000-row sample of A as the spline model. River 0.22.0's adaptive random forest uses ten trees and default hyperparameters apart from its seed [6, 27]. It processes the same initial A sample in chronological order, then learns each delayed queried label once. We report the forest with its natural 0.5 threshold, its initial B threshold, and recent thresholds. Its accumulated tree state and reuse of calibration observations for learning distinguish it from the disjoint 192/64 head experiment. It is an equal-acquisition-budget benchmark, not a claim of equal total memory or an isolated head intervention.

To expose operating compromises, each relevant score trajectory also selects thresholds minimizing

$$L_\lambda(\tau)=\mathrm{FNR}(\tau)+\lambda\,\mathrm{FPR}(\tau),\qquad \lambda\in\{0.25,0.5,1,2,4\}. \tag{7}$$

Each threshold is estimated from the controller's available past calibration labels and applied to future blocks. B supplies the initial threshold for the same objective. $\lambda=1$ reproduces balanced-accuracy selection. The plotted points are the outcomes of these causal policies; they are not an oracle ROC envelope selected from D labels. For prevalence π and per-error costs $C_{\mathrm{FN}}, C_{\mathrm{FP}}$, the corresponding rate weight is $\lambda=C_{\mathrm{FP}}(1-\pi)/(C_{\mathrm{FN}}\pi)$. Thus the grid represents explicit relative rate costs rather than a universal operational utility.

All five spline seeds additionally run on TON D with delays 0, 5, and 20 blocks, and with a 0.1% budget at delay one. No setting replaces the original primary analysis after seeing its outcome. The salt and age/minimum rules stay fixed, allowing these tests to expose label starvation instead of changing the controller to suppress it.

3.8 Separate chronological evaluation on UNSW-NB15

The extension uses the four full UNSW-NB15 network CSVs from a versioned public mirror, with SHA-256 hashes recorded for each file. Three exact repetitions at consecutive file boundaries are verified against the complete raw rows and removed; this recovers the provider's 2,540,044 records. There is no general deduplication. The selected features are `dur`, `sbytes`, `dbytes`, `sloss`, `dloss`, `Spkts`, `Dpkts`, and `tcprtt`. They are compact numerical flow descriptors, but their semantics are not identical to the TON feature set.

Records are sorted by $\max(\mathrm{Ltime}, \mathrm{Stime}+\mathrm{float32}(\mathrm{dur}))$, then stable row identity. The maximum respects both the recorded last-packet time and duration; the archive's integer-second timestamps limit temporal precision. The valid ordered rows are split by fixed fractions: 40% A (1,016,017 rows), 20% B (508,009), and 40% D (1,016,018). Equal timestamps occur at partition boundaries and are ordered by row identity; there is no temporal embargo or claim of event independence. These fractions and all model settings were specified before predictive results. Models are trained anew; the publisher's preselected train/test tables are unused. The D interval contains 803,736 normal and 212,282 attack records. Its A/B-unseen stratum has 523,155 rows, including 483,783 normal and 39,372 attack records. The mask affects reporting only.

3.9 Additional summaries and uncertainty

We retain per-block confusion counts and class counts, report attack-family recall, and add pooled AUROC for fixed initial scores and normal-count-weighted block AUROC. Adaptive scores are not pooled across different model states. To describe the original factorial design, the balanced-accuracy interaction is

$$I_{\mathrm{BA}} = (\mathrm{BA}_{\mathrm{FIT,CAL}} - \mathrm{BA}_{\mathrm{FIT,B}}) - (\mathrm{BA}_{\mathrm{Frozen,CAL}} - \mathrm{BA}_{\mathrm{Frozen,B}}). \tag{8}$$

This is a contrast of the four implemented policies, not a model-free causal percentage of adaptation benefit.

Five seeds describe algorithmic randomization conditional on a capture. We do not use a nominal sign-test p-value as evidence across independent streams, especially given the original sampler's seed/block reuse. Supplementary moving-block resampling uses lengths 5, 20, and 50 and 2,000 replicates of recorded paired error counts. These are descriptive sensitivity ranges: replay states are not refitted after resampling, and nonstationarity precludes interpreting them as generalization confidence intervals. The separate UNSW capture supplies a more direct check of transfer of the empirical pattern.

4 Original TON_IoT campaign results

4.1 Development evidence motivating the final comparison

Development on C proceeded through four recorded stages. These stages answer different questions and are not a single ranking of competing algorithms. In the initial adaptation study, selecting updates using evidence of score inversion outperformed a random schedule with the same number of updates in mean block AUROC by 0.0217, 0.0559, and 0.0719 for logistic regression, an MLP, and the additive spline model, respectively, against the mean of 20 random schedules. For the additive spline model, none of these update policies exceeded the frozen mean block AUROC of 0.8911: evidence-driven, continuous, and random updates attained 0.8341, 0.8073, and 0.7621. The fitted-memory inversion diagnostic was in sample, and out-of-sample evidence from requested block labels was often unavailable or weak after the required delay. Improved ranking did not consistently improve both error counts.

The next stage changed the memory policy while retaining a 256-example capacity. For the continuously updated spline model, class-aware memory increased mean block AUROC by about 0.084 compared with global FIFO, while attack recall fell by 56.66 percentage points at the fixed historical threshold. Subsequent threshold experiments showed why score and decision changes had to be separated. Re-selecting only the frozen spline threshold improved recall on C by 17.73 percentage points and increased FPR by 1.48 percentage points, with exactly unchanged AUROC. Recalibration using the entire B archive was also explored during development, but that configuration required the additional historical archive and is not presented as satisfying the final 256-example online-memory contract.

The final development configuration separated FIT192 and CAL64. For seed 42 on C, FIT192 / CAL64 reduced FPR by 16.08 percentage points relative to Frozen / CAL64, but reduced recall by 2.60 percentage points and mean block AUROC by about 0.109. This mixed outcome motivated preserving the four traces and reporting FN and FP separately in the final evaluation. The C findings are development evidence, not an additional independent confirmation of the final D result.

Table 4: Final rates: mean ± sample SD over five paired seeds. Recall, FPR, and BA are percentages calculated globally. AUROC is the unweighted mean over valid blocks (352/355 for all D; 350/355 for A/B/C-unseen vectors), then averaged over seeds.

Trace	Recall (%)	FPR (%)	BA (%)	Mean block AUROC
All D				
Frozen / B	21.11 ± 0.72	12.64 ± 0.13	54.24 ± 0.33	0.8536 ± 0.0122
Frozen / CAL64	89.09 ± 1.64	46.74 ± 2.04	71.18 ± 1.73	0.8536 ± 0.0122
FIT192 / B	8.81 ± 5.89	0.88 ± 1.00	53.96 ± 3.23	0.9359 ± 0.0034
FIT192 / CAL64	89.38 ± 1.53	20.47 ± 2.11	84.46 ± 1.15	0.9359 ± 0.0034
Vectors not observed in A/B/C (same replay)				
Frozen / B	3.12 ± 1.11	5.30 ± 0.20	48.91 ± 0.52	0.7934 ± 0.0229
Frozen / CAL64	87.83 ± 3.05	33.99 ± 3.25	76.92 ± 2.82	0.7934 ± 0.0229
FIT192 / B	1.47 ± 0.27	0.71 ± 0.18	50.38 ± 0.19	0.9051 ± 0.0049
FIT192 / CAL64	91.20 ± 1.83	31.94 ± 1.11	79.63 ± 1.01	0.9051 ± 0.0049

Table 5: Absolute final error counts: mean ± sample SD over five seeds. Noninteger means result from averaging integer counts. The paired differences in Table 6 retain the seed correspondence.

Trace	False negatives	False positives
All D		
Frozen / B	$2{,}556{,}210.8 \pm 23{,}236.3$	$39{,}136.2 \pm 413.1$
Frozen / CAL64	$353{,}454.0 \pm 53{,}161.4$	$144{,}730.2 \pm 6{,}318.6$
FIT192 / B	$2{,}954{,}958.6 \pm 190{,}791.8$	$2{,}737.2 \pm 3{,}083.7$
FIT192 / CAL64	$344{,}198.8 \pm 49{,}694.1$	$63{,}374.8 \pm 6{,}542.7$
Vectors not observed in A/B/C (same replay)		
Frozen / B	$2{,}032{,}985.0 \pm 23{,}235.3$	$6{,}887.2 \pm 260.8$
Frozen / CAL64	$255{,}423.0 \pm 63{,}982.9$	$44{,}176.4 \pm 4{,}223.8$
FIT192 / B	$2{,}067{,}609.4 \pm 5{,}672.1$	917.6 ± 228.5
FIT192 / CAL64	$184{,}594.4 \pm 38{,}339.5$	$41{,}521.2 \pm 1{,}447.6$

4.2 Absolute outcomes on the final interval

Tables 4 and 5 report all four traces, including the controls that retain the B threshold. With frozen weights, recent threshold selection raises mean attack recall on all D from 21.11% to 89.09%, while FPR rises from 12.64% to 46.74%. Mean block AUROC remains 0.8536 because the scores are identical. Thus, a large recovery in recall can arise entirely from moving the decision threshold.

The FIT192/B diagnostic displays the expected risk of retaining a threshold after changing the fitted sample and score scale. FIT192 / B reaches mean block AUROC 0.9359, yet attack recall is only 8.81% and FPR is 0.88%. The ranking improves on average, but the inherited threshold misses most attacks. Refitting with approximately balanced class quotas differs from fitting A, where about 94% of observations are attacks. Under pure prior shift, the log-odds correction would be an intercept shift; here reweighted slopes and changing class-conditional traffic also contribute. Thus this control illustrates score–threshold mismatch rather than a newly discovered property of logistic models. With the same adapted scores and a recent threshold, FIT192 / CAL64 reaches recall 89.38% and FPR 20.47%. Relative to FIT192 / B, this threshold change removes 2,610,759.8 mean false negatives at the cost of 60,637.6 additional false positives. Figure 2 shows the resulting operating-point changes.

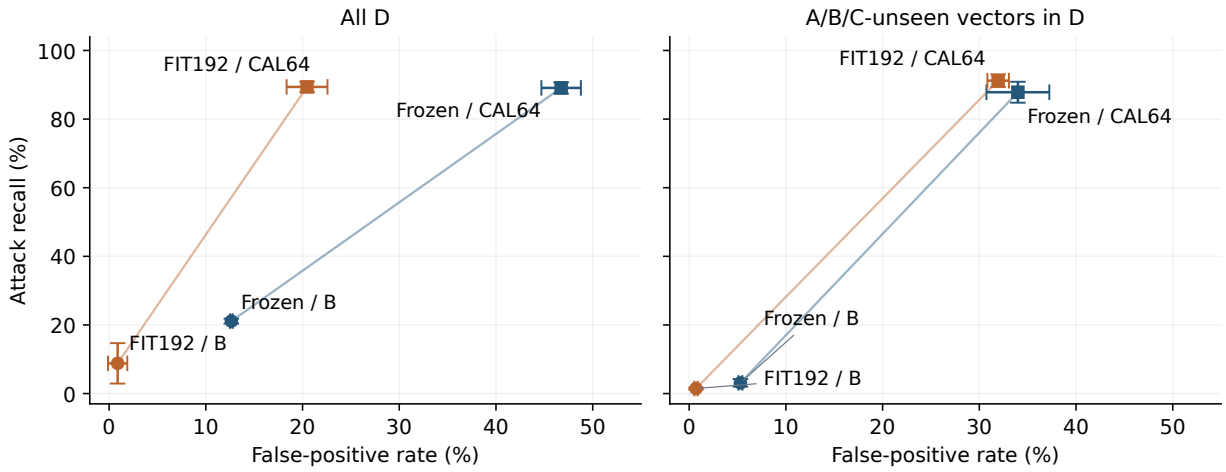


Figure 2: Operating points on the same final replay. Points are seed means; bars show sample SD. Lines connect traces with identical score functions and different threshold policies, not a continuously optimized ROC curve. The large threshold effects coexist with a narrower primary comparison between the two CAL64 traces.

Table 6: Primary paired differences, FIT192 / CAL64 minus Frozen / CAL64. Negative values indicate fewer errors. Seeds 43, 45, and 46 jointly improve FN and FP in both reporting scopes; seed 44 worsens both on A/B/C-unseen vectors.

	All D		A/B/C-unseen vectors	
Seed	ΔFN	ΔFP	ΔFN	ΔFP
42	+54,175	−84,858	+4,271	−3,659
43	−115,380	−89,311	−165,016	−6,888
44	+125,148	−73,762	+59,443	+3,547
45	−40,935	−77,720	−129,600	−4,684
46	−69,284	−81,126	−123,241	−1,592

4.3 Primary paired comparison and the joint-error criterion

The primary contrast compares FIT192 / CAL64 with Frozen / CAL64. On all D, mean balanced accuracy improves by 13.28 ± 2.26 percentage points and mean block AUROC by 0.0823 ± 0.0155. Mean FP decreases by $81{,}355.4 \pm 6{,}051.6$. Mean FN decreases by $9{,}255.2 \pm 97{,}413.0$, corresponding to a recall difference of 0.29 ± 3.01 percentage points. Hence the average balanced-accuracy gain is predominantly associated with lower false-positive rates, while the missed-attack outcome is variable.

The per-seed counts in Table 6 and Figure 3 show the limitation directly. All five seeds reduce FP on the full stream, but seeds 42 and 44 increase FN. Seeds 43, 45, and 46 improve both counts. The pre-specified four-of-five criterion is therefore not met. The sign of the mean FN change does not overturn that result, and the sample SD should not be read as a confidence bound supporting superiority.

4.4 Previously unobserved vectors and concentration of the FP gain

For D vectors absent from A/B/C, FIT192 / CAL64 reaches 91.20% attack recall but an FPR of 31.94%. The corresponding Frozen / CAL64 rates are 87.83% and 33.99%. The paired BA gain is 2.71 ± 3.63 percentage points, substantially smaller than the full-stream gain. Mean FN decreases by $70{,}828.6 \pm 97{,}060.6$ and mean FP by $2{,}655.2 \pm 3{,}957.2$. Again, only three of five seeds jointly

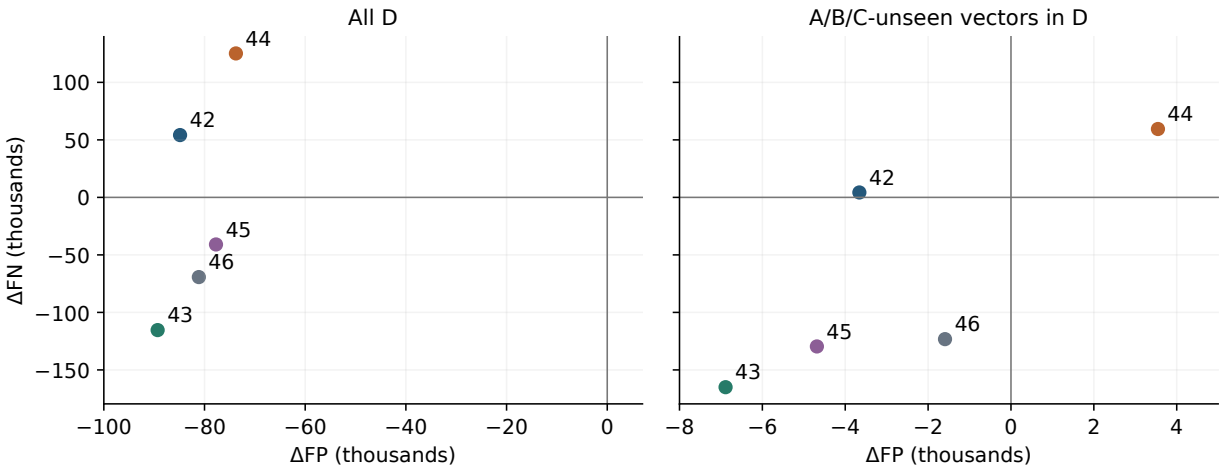


Figure 3: Primary paired differences by seed. Both axes show thousands of errors. Joint improvement requires a point in the lower-left quadrant (including a non-deteriorating boundary). The horizontal scales differ to make the much smaller unseen-vector FP changes visible.

Table 7: Complementary overlap strata in the original campaign: mean ± seed SD. Rates are percentages. The same adaptive trajectory supplies both strata.

Stratum	Trace	Recall	FPR	BA
Seen	Frozen / CAL64	91.41 ± 2.00	55.96 ± 2.11	67.73 ± 1.22
Seen	FIT192 / CAL64	86.02 ± 2.30	12.16 ± 3.72	86.93 ± 2.49
Unseen	Frozen / CAL64	87.83 ± 3.05	33.99 ± 3.25	76.92 ± 2.82
Unseen	FIT192 / CAL64	91.20 ± 1.83	31.94 ± 1.11	79.63 ± 1.01

improve both counts.

The complementary seen stratum exposes the trade-off (Table 7). Head adaptation lowers recall from 91.41% to 86.02%, while FPR falls from 55.96% to 12.16%. It adds 61,573.4 mean FN while removing 78,700.2 FP. In the unseen stratum it removes 70,828.6 FN and 2,655.2 FP. Thus the small full-stream mean FN reduction combines effects with opposite signs. The seen stratum contributes 96.7% of the total FP reduction, but that percentage describes the location of FP savings rather than the fraction of all benefit caused by overlap. The strata differ in traffic composition and share one adaptive state.

The factorial BA interaction is positive in every original full-stream seed: 10.76–17.65 percentage points, with mean 13.55. On unseen vectors it ranges from −4.43 to +4.89 points, with mean 1.24. The gain from jointly changing the head and threshold therefore depends on the reporting population. The four-trace interaction and the complementary-stratum decomposition reveal information concealed by a single average BA gain.

4.5 When limited labels prevent adaptation

Label availability creates a distinct temporal failure mode in the C development replay. There are 440 skipped update decisions, all associated with insufficient recent normal examples for calibration. The longest consecutive sequence spans decision blocks 241–663 and affects predictions in blocks 242–664: 423 blocks. Figure 4 shows the eligible normal-label count and the corresponding decision rates.

Stratifying predictions by the preceding update decision yields opposite FN patterns. Across 474 blocks following applied updates, FIT192 / CAL64 has 29,697 fewer FN and 26,528 fewer FP than

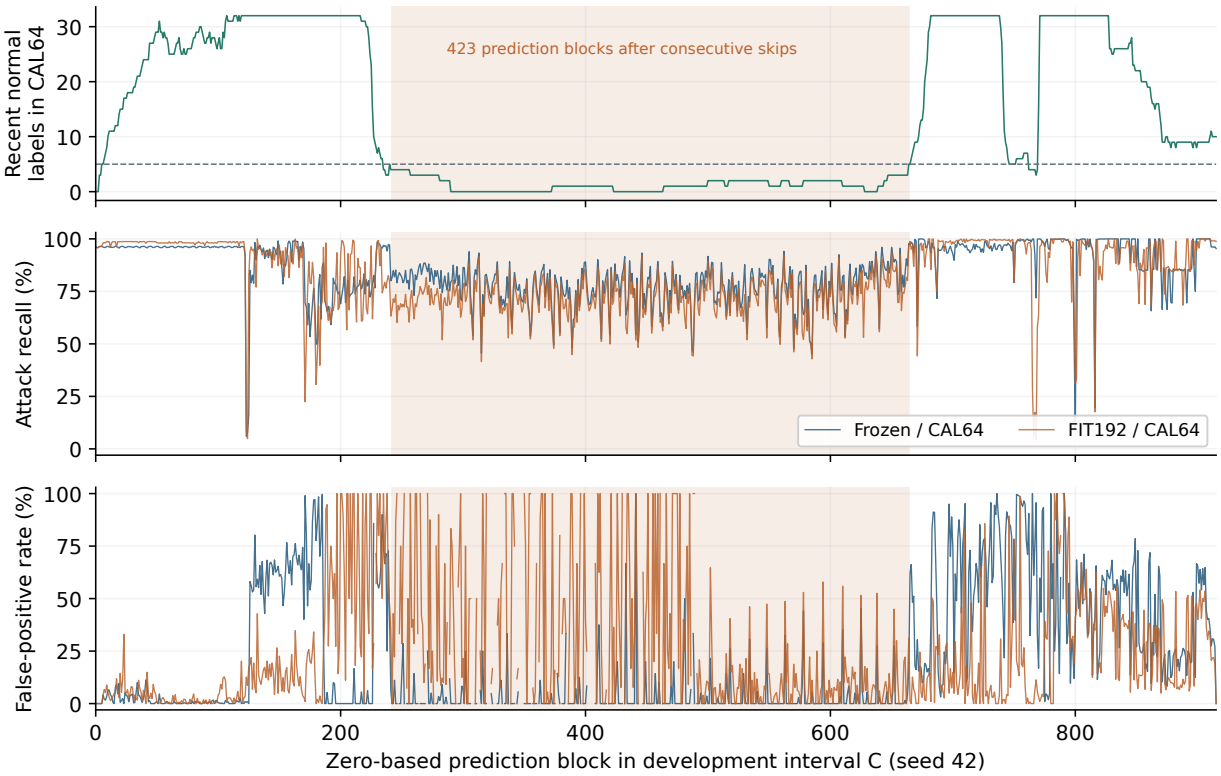


Figure 4: Development interval C, seed 42. Shading marks prediction blocks 242–664, following 423 consecutive skipped update decisions. The upper panel shows the recorded number of recent normal examples in CAL64, with the minimum of five indicated. Recall and FPR are per-block values; undefined class-specific rates remain gaps. The display is descriptive and does not randomize the occurrence of skips.

Frozen / CAL64. Across 440 blocks following skips, it has 263,032 more FN and 1,342 fewer FP. In the longest skipped segment alone, both counts worsen: FN increases by 219,608 and FP by 807. These are post hoc associations conditioned on the realized stream. The segments have different class composition and cannot establish that skipping caused the deterioration.

The final D replay behaves differently. Of 354 recorded update opportunities per seed, 337, 351, 347, 341, and 350 are accepted for seeds 42–46, respectively. Their 17, 3, 7, 13, and 4 skips all occur before the first valid update. There is no recurrence of the long post-update starvation interval observed on C. Thus, the D evaluation confirms the four-trace outcome on a later stream, but it is not a second observation of the same long-skip phenomenon.

4.6 Measured computation and its boundary

Table 8 gives recorded computational costs on the original evaluation machine: Windows 11, Python 3.13.2, NumPy 2.3.4, scikit-learn 1.7.2, an Intel processor reported as Family 6 Model 186 Stepping 3, 12 visible CPU threads, and 16,067.4 MiB of system RAM. These are elapsed times from one execution environment, not cross-device benchmarks.

The five-seed command takes 1,234.388 seconds within its recorded timer boundary. This includes D hashing and loading, initial A fitting and B preparation, the five D replays, and writing the five per-seed reports. It excludes the earlier A/B/C hash calculation and final summary serialization. Within each replay, accumulated head-fitting time is 9.630–10.672 seconds; each of the two accumulated threshold-selection components is below 0.11 seconds. These observations support only the relative cost of these implementation components in this run. They do not establish an energy advantage, a device-level latency guarantee, or an embedded memory footprint.

Table 8: Recorded final-run costs in seconds. Fits and threshold-selection times are components included within replay time, not extra times to add to it. The final recorded update has no subsequent evaluated block.

Seed	Updates	Skips	Head fit	Frozen threshold	Adapted threshold	D replay
42	337	17	10.672	0.103	0.059	59.837
43	351	3	10.224	0.090	0.052	52.683
44	347	7	9.630	0.083	0.050	49.032
45	341	13	9.632	0.094	0.055	48.622
46	350	4	10.378	0.090	0.057	53.348

5 Post-review experiments

5.1 Equal acquisition budgets expose a different FN/FP compromise

Table 9 reports the new TON runs with independent seed/block randomization. The spline head with CAL64 attains 89.46% recall and 19.78% FPR. A frozen spline detector using all acquired labels through CAL256 attains higher recall, 92.38%, with 48.14% FPR. The head therefore trades 2.92 recall points for a 28.36-point FPR reduction; its BA gain is 12.72 points. In counts, the mean difference is +94,549.4 FN and −87,825.2 FP. All five seeds incur more FN and fewer FP. This is a consistent trade-off, not joint error dominance.

The stronger threshold-only baseline changes the interpretation of the residual gain. Against CAL256, the spline head adds 75,144.6 FN and removes 79,459.4 FP in the seen stratum. On unseen vectors it adds 19,404.8 FN and removes 8,365.8 FP on average; two seeds improve both error counts there. Thus 90.5% of the FP reduction is still concentrated on previously observed vectors, but the unseen recall improvement observed against the original CAL64 control is no longer a general finding. Figure 5 shows the paired rate differences. The control used to define "additional benefit" is consequential.

The linear head reaches 84.80% BA, close to the spline head's 84.84%, with 88.96% recall and 19.36% FPR. Its frozen/CAL256 control reaches 87.20% recall and 43.10% FPR. The allocation of gain between recall and false alarms is therefore model-dependent even on the same requested labels.

The standard stream learner changes the picture further. ARF at its natural threshold 0.5

Table 9: Post-review TON D results, five seeds: mean ± sample SD. Recall, FPR, and BA are percentages. The last column is macro block AUROC. All methods receive the same acquired labels; their use of those labels and model memory differs as specified in Table 3.

Controller	Recall	FPR	BA	AUROC
Spline: Frozen / B	20.93 ± 0.37	12.59 ± 0.14	54.17 ± 0.18	0.850 ± 0.017
Spline: Frozen / CAL64	91.37 ± 2.24	47.93 ± 4.68	71.72 ± 1.73	0.850 ± 0.017
Spline: Frozen / CAL256	92.38 ± 1.32	48.14 ± 4.67	72.12 ± 2.42	0.850 ± 0.017
Spline: FIT192 / CAL64	89.46 ± 0.89	19.78 ± 0.77	84.84 ± 0.79	0.937 ± 0.004
LR: Frozen / B	27.00 ± 1.22	76.01 ± 1.45	25.50 ± 0.21	0.432 ± 0.005
LR: Frozen / CAL256	87.20 ± 2.72	43.10 ± 12.33	72.05 ± 5.83	0.432 ± 0.005
LR: FIT192 / CAL64	88.96 ± 1.49	19.36 ± 2.42	84.80 ± 1.84	0.874 ± 0.009
ARF: Frozen / B	16.46 ± 7.19	34.40 ± 17.63	41.03 ± 8.76	0.540 ± 0.073
ARF: Frozen / CAL256	76.32 ± 15.88	40.57 ± 8.27	67.87 ± 8.19	0.540 ± 0.073
ARF: threshold 0.5	97.98 ± 0.15	19.03 ± 2.94	89.47 ± 1.47	0.957 ± 0.005
ARF: threshold B	61.38 ± 30.60	0.31 ± 0.14	80.54 ± 15.25	0.957 ± 0.005
ARF: CAL64	95.95 ± 1.11	20.92 ± 3.05	87.52 ± 1.59	0.957 ± 0.005
ARF: CAL256	96.41 ± 0.64	15.62 ± 1.94	90.39 ± 0.97	0.957 ± 0.005

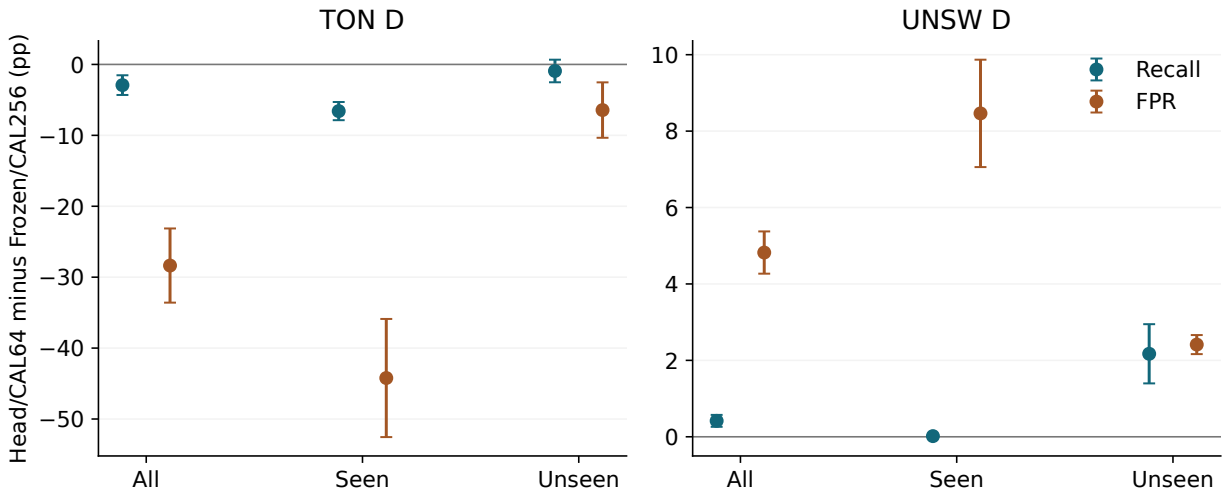


Figure 5: Spline FIT192/CAL64 minus Frozen/CAL256, by reporting stratum. Points are paired seed means; bars are sample SDs. Positive recall differences and negative FPR differences favor the head. When both rates move in the same direction, the policy trades recall against false alarms.

Table 10: Spline controls in the new campaign, five seeds. Rates are mean ± sample SD in percent. Independent CAL64 equals the matched CAL64 outcomes in these runs.

Dataset	Controller	Recall	FPR	BA
TON	Independent CAL64	91.37 ± 2.24	47.93 ± 4.68	71.72 ± 1.73
TON	Offset / B	11.51 ± 7.70	1.70 ± 1.19	54.91 ± 3.49
TON	Offset / CAL64	91.37 ± 2.24	47.93 ± 4.68	71.72 ± 1.73
TON	Prior-centred / CAL64	89.59 ± 1.18	20.74 ± 1.98	84.42 ± 1.13
UNSW	Independent CAL64	96.63 ± 0.25	9.12 ± 0.72	93.75 ± 0.34
UNSW	Offset / B	97.23 ± 0.14	20.42 ± 0.10	88.41 ± 0.04
UNSW	Offset / CAL64	96.63 ± 0.25	9.12 ± 0.72	93.75 ± 0.34
UNSW	Prior-centred / CAL64	96.88 ± 0.15	9.43 ± 0.64	93.72 ± 0.34

reaches 97.98% recall, 19.03% FPR and 89.47% BA; ARF/CAL256 reaches 96.41%, 15.62% and 90.39%, respectively. Its own frozen/CAL256 control reaches only 67.87% BA. ARF/CAL256 reduces both error counts relative to that control in every TON seed. This is substantial learning benefit beyond threshold adaptation for the tested forest, under its more permissive model-memory and calibration-reuse contract. It also prevents the spline-head findings from being generalized to all adaptive IDS learners.

5.2 Which proposed mechanisms survive direct controls?

Independently updating CAL64 produces the same decisions as the matched-gate control in both datasets' main runs. Removing the head gate alone therefore does not explain the large original FP contrast. Both versions still require enough recent calibration observations; on development C, the recorded starvation was caused by missing normal CAL labels, so this change alone would not remove that eligibility condition.

The intercept-only/CAL64 trace exactly matches frozen/CAL64 confusion counts in every new spline and linear run, as expected from threshold-shift invariance. Its fixed-B trace can move recall markedly, but that demonstrates an operating-point change rather than new discrimination. Prior-centred regularization also does not supply a uniform remedy. For the spline model, it yields 84.42% BA on TON versus 84.84% for the zero-centred refit, and 93.72% on UNSW versus 93.22%. The regularization centre affects the result, but the tested alternative does not establish a generally

Table 11: Separate chronological UNSW-NB15 D evaluation, five seeds, with the same reporting conventions as Table 9. Each model is trained anew on that dataset's A partition.

Controller	Recall	FPR	BA	AUROC
Spline: Frozen / B	94.53 ± 0.11	1.95 ± 0.02	96.29 ± 0.05	0.977 ± 0.000
Spline: Frozen / CAL64	96.63 ± 0.25	9.12 ± 0.72	93.75 ± 0.34	0.977 ± 0.000
Spline: Frozen / CAL256	96.37 ± 0.07	5.53 ± 0.12	95.42 ± 0.06	0.977 ± 0.000
Spline: FIT192 / CAL64	96.79 ± 0.14	10.36 ± 0.50	93.22 ± 0.22	0.970 ± 0.003
LR: Frozen / B	90.16 ± 4.91	6.33 ± 8.87	91.92 ± 2.00	0.921 ± 0.080
LR: Frozen / CAL256	93.82 ± 2.82	10.09 ± 7.18	91.86 ± 2.20	0.921 ± 0.080
LR: FIT192 / CAL64	94.97 ± 0.48	12.20 ± 0.64	91.39 ± 0.49	0.944 ± 0.005
ARF: Frozen / B	47.21 ± 48.03	25.48 ± 42.53	60.87 ± 15.86	0.505 ± 0.220
ARF: Frozen / CAL256	47.57 ± 47.74	26.09 ± 42.15	60.74 ± 15.90	0.505 ± 0.220
ARF: threshold 0.5	86.91 ± 1.65	2.09 ± 0.42	92.41 ± 0.65	0.978 ± 0.004
ARF: threshold B	98.98 ± 0.93	97.73 ± 1.86	50.63 ± 0.65	0.978 ± 0.004
ARF: CAL64	97.21 ± 0.92	8.53 ± 1.62	94.34 ± 0.86	0.978 ± 0.004
ARF: CAL256	96.48 ± 1.03	4.77 ± 0.76	95.86 ± 0.38	0.978 ± 0.004

preferable update rule (Table 10).

5.3 A separate capture reverses the aggregate preference

On UNSW-NB15, the frozen spline model at its initial B threshold already reaches 94.53% recall, 1.95% FPR, and 96.29% BA (Table 11). Recent CAL256 selection raises recall to 96.37%, but raises FPR to 5.53%. Adding the head gives 96.79% recall and 10.36% FPR, reducing BA to 93.22%. Against CAL256, the head removes 885.0 FN while adding 38,754.6 FP; every seed has that same directional trade-off. On unseen vectors its mean recall gain is 2.17 points and its FPR increase is 2.41 points, leaving a mean BA difference of -0.12 points.

The linear model also illustrates why the baseline matters: its head/CAL64 BA is 91.39%, below frozen/CAL256 at 91.86%, although it exceeds frozen/CAL64 at 89.75%. A small calibration memory can make adaptation appear more beneficial than it is against a controller using all acquired labels. The second capture therefore supports the need for decomposition, while contradicting a universal claim that the specific adaptive head improves detection.

ARF/CAL256 reaches 96.49% recall, 4.77% FPR and 95.86% BA on UNSW. Its BA exceeds its own frozen/CAL256 baseline in every seed; the latter is highly variable across initial fits and averages 60.74% BA. Nevertheless, none of these paired UNSW seeds improves both FN and FP at once. Even for a strong stream learner, mean error reductions and consistent BA gains must be distinguished from per-seed joint dominance. The frozen spline/B detector retains the highest mean BA among the displayed default policies on this capture, at 96.29%.

5.4 Causally selected operating points and label sensitivity

Figure 6 expresses the practical question as a family of explicit error compromises. It prevents the BA optimum from being mistaken for a unique preferred detector. The error-cost weight, available labels, and score function jointly determine which future operating points are reached. A preferred point consequently requires an application-specific alarm burden and missed-attack cost.

Delay and label scarcity materially change those points (Table 12). For the spline head on TON, BA decreases from 85.90% at delay zero to 84.84%, 82.87%, and 80.07% at delays one, five, and twenty. At a 0.1% budget and delay one, recall falls to 32.13% and BA to 57.49%; the frozen/CAL256 controller retains 86.22% recall and 79.48% BA. This is a consequence of the tested finite-memory and eligibility policy under fewer labels, not evidence that every adaptive learner has the same sensitivity.

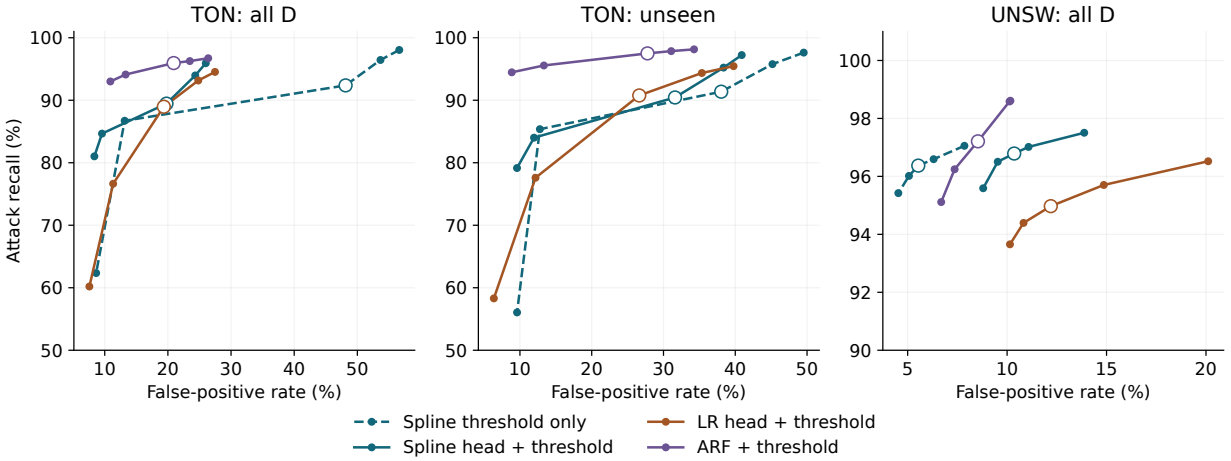


Figure 6: Operating points generated by five pre-specified rate-cost weights, each threshold fitted only to past available labels. Connected points belong to one controller; hollow markers denote $\lambda = 1$. Vertical ranges differ to expose the observed variation. The display compares realized future errors, rather than an oracle frontier optimized on evaluation labels. Means are over five paired seeds; individual results are supplied in the accompanying CSV tables.

Table 12: TON spline sensitivity, five seeds. The default budget is 1%; only the last pair uses 0.1%. Rates are percentages. Updates count eligible decisions, including the final decision without a subsequent evaluated block.

Condition	Controller	Recall	FPR	BA	Updates
Delay 0	Frozen/CAL256	92.69 ± 1.33	46.24 ± 4.93	73.23 ± 2.52	353.2 ± 0.4
Delay 0	FIT192/CAL64	89.62 ± 0.95	17.82 ± 1.02	85.90 ± 0.89	348.8 ± 2.6
Delay 1	Frozen/CAL256	92.38 ± 1.32	48.14 ± 4.67	72.12 ± 2.42	352.2 ± 0.4
Delay 1	FIT192/CAL64	89.46 ± 0.89	19.78 ± 0.77	84.84 ± 0.79	347.8 ± 2.6
Delay 5	Frozen/CAL256	91.22 ± 1.30	46.72 ± 7.96	72.25 ± 4.07	348.2 ± 0.4
Delay 5	FIT192/CAL64	87.96 ± 0.79	22.22 ± 4.11	82.87 ± 1.78	343.8 ± 2.6
Delay 20	Frozen/CAL256	87.07 ± 1.25	37.49 ± 6.06	74.79 ± 3.32	333.2 ± 0.4
Delay 20	FIT192/CAL64	83.32 ± 0.89	23.18 ± 6.65	80.07 ± 3.09	322.4 ± 7.7
Budget 0.1%	Frozen/CAL256	86.22 ± 4.35	27.25 ± 4.71	79.48 ± 4.02	292.4 ± 34.6
Budget 0.1%	FIT192/CAL64	32.13 ± 24.30	17.14 ± 9.81	57.49 ± 7.35	85.6 ± 7.0

The nominal 1% budget corresponds to 35,499 queried labels during 62.77 observed hours of TON D, approximately 566 labels per hour. UNSW D requires 10,160 labels during 7.89 hours, approximately 1,288 per hour. These volumes describe an annotation service assumption rather than a single analyst's feasible workload. Measured query-to-arrival times have medians near 3.50 minutes on TON and 6.45 minutes on UNSW; their approximate 95th percentiles are 54.6 and 12.5 minutes, respectively. The exact per-seed distributions are provided with the results. A one-block delay is therefore neither a constant number of seconds nor a guarantee of realistic incident-confirmation latency.

5.5 Ranking summaries, attack families, and computational accounting

Pooled fixed-score AUROC and two block summaries are reported in Table 14; per-family attack recall appears in Table 15. These quantities answer different questions. In particular, macro block AUROC gives a block with few normal observations the same weight as a block with many. Normal-count weighting is a sensitivity summary, while pooling the fixed score also compares observations across time. Neither is substituted silently for the original statistic.

All additional runs record initial fitting, initial B scoring, cached feature/score preparation, replay components, and serialization sizes. The initial artifact contains the base model and transform;

the final size counts only the base model object. Both exclude label memories, the pending queue, inference caches and separately held linear-head vectors, and neither measures total controller RAM. The extension reuses immutable representations and identical-vector predictions within a fixed forest state to accelerate the offline study. Consequently, cached replay time alone is not an end-to-end inference benchmark. The complete cost ledger reports preparation separately, and the forest's persistent tree state is not described as satisfying a 256-example total-memory bound. No embedded latency or energy claim follows from these measurements.

6 Discussion

6.1 The relevant result is conditional incremental benefit

The original TON campaign answers the question that motivated the study: a large recall recovery can be produced by moving a frozen detector's threshold, while the refitted head primarily changes false alarms. Yet the extension shows why this statement needs a specified comparator. Against a threshold fitted using all acquired labels, the spline head consistently sacrifices recall to reduce FP. On the separately collected UNSW capture, the same head instead increases recall slightly while increasing FP enough to lower BA. The useful finding is the location and direction of these conditional error changes, not a universal percentage of gain attributable to adaptation.

Even a numerical "share explained by the threshold" is path-dependent in the four-trace design. The contrast Frozen/B to Frozen/CAL64 holds scores fixed, whereas FIT/B to FIT/CAL64 operates on a different score trajectory. Their BA interaction is large on full TON D and small on unseen vectors. Reporting both paths and the stronger CAL256 comparison is more informative than assigning one model-free percentage to threshold and weight learning.

The seen/unseen decomposition adds a second qualification. In the original experiment, the average full-stream FN difference combines a recall loss on seen vectors with a gain on unseen vectors. In the stronger new comparison, both strata incur more FN on average. These are genuine changes in the error allocation induced by the tested policies. They do not establish memorization or prove that prior overlap caused the benefit: feature collisions, different class mixtures, and one shared adaptive trajectory remain possible contributors.

6.2 A threshold is part of the adaptive system

The FIT192/B trace should be understood as an intentionally diagnostic operating-point mismatch. Refitting on a class-balanced memory changes both the effective class prior and the fitted coefficients, so retaining a threshold chosen for the original score is not a strong deployment policy. The offset control and freely reselected threshold establish an exact invariance: an intercept change alone supplies no additional attainable decisions. A meaningful learning advantage must therefore appear beyond that adjustment and beyond a threshold controller with fair access to the acquired labels.

Label eligibility is equally consequential. Independent CAL64 does not change the new main outcomes because its calibration requirement remains the limiting condition. CAL256 can use more of the acquired information and becomes particularly competitive when the query budget decreases. The low-budget and long-delay results suggest that an adaptive system should expose its recent per-class evidence and update status, rather than treating a configured annotation percentage as proof that useful updates are possible.

6.3 Interpreting operational costs

For attack prevalence π, holding recall r and false-positive rate f fixed gives

$$\text{precision}(\pi) = \frac{\pi r}{\pi r + (1-\pi) f}. \tag{9}$$

Using the new TON spline-head rates gives approximately 4.37% precision at 1% attack prevalence and 0.45% at 0.1%. These are illustrative base-rate calculations, conditional on class-conditional performance transferring to the hypothetical population. They are not deployment predictions. The same calculation can be performed for every supplied operating point, together with an explicitly stated traffic rate to obtain alerts per hour. Without that traffic rate and a justified cost ratio, BA does not identify an operational optimum.

The annotation load is also substantial even at 1%. Reducing it by a factor of ten causes the tested head controller to lose much of its recall. Acquisition, confirmation delay, model updates, and alarm review must therefore be accounted for together. Our experiment measures algorithmic error compromises under a declared feedback service; an operational study would additionally need reliable label arrival distributions and analyst costs.

6.4 A reusable evaluation checklist

The following reporting obligations follow directly from the comparisons:

1. Preserve the four combinations of frozen or adapted scores and historical or recent thresholds.
2. Match acquired labels and delay, then disclose effective label use, memory retention, and whether fitting and threshold data overlap; include a threshold-only control using all acquired labels.
3. Keep prediction before feedback, log query identities and earliest influence times, and report skipped updates with recent class counts.
4. Report global FN/FP, class-conditional rates, ranking summaries with their denominators, and operating points selected only from past evidence.
5. Separate row chronology, feature-vector overlap, attack families, and independent captures; retain unfavorable strata and previously specified acceptance criteria.

This checklist does not prescribe a particular architecture or a single acceptance metric. It makes the claim being evaluated and the source of any observed improvement explicit.

6.5 Threats to validity

Evidence stages and dependence. The original TON procedure was fixed before its predictive campaign, but D had already been inspected structurally. The extension uses known TON outcomes to motivate new questions and is explicitly post hoc on that capture. UNSW provides a separate laboratory capture, not five independent environments. UNSW had appeared in companion project studies; the chronological protocol here is new, but the dataset is not an untouched resource in the broader project history. Five seeds quantify initialization and acquisition randomization; their SDs and the descriptive moving-block ranges do not establish uncertainty across operational networks. The role salt is fixed across seeds.

Historical exposure and representation. An earlier project sample contains eight-feature matches to 889,093 TON D rows (25.05%), separately from A/B/C overlap. The initial models here are refitted on A. During revision, the historical 211,043-row CSV was recovered from a public copy with the recorded SHA-256, closing the former missing-copy provenance issue; historical exposure itself remains. Exact equality in eight numerical features neither identifies an event nor detects approximate, session, or host dependence. The feature contracts omit many potentially informative

fields, and the fixed spline functions cannot learn new interactions or extrapolate beyond their clipped ranges.

Feedback and operating assumptions. Dataset labels are treated as correct. Analyst errors, adversarial feedback, and asynchronous individual label delays are outside the replay. Per-family reporting reduces binary aggregation but small families still support limited precision. The forest uses more persistent model state and reuses calibration labels in training, whereas the head experiment enforces disjoint roles. Equal query budgets must not be read as equal computational resources or equal estimator independence.

Numerical reproducibility. The extension verifies all 23 TON source-file hashes, partition counts, overlap counts, and 710 original D block digests for row identity and labels. Rebuilding A through the unmodified original parser gives byte-identical output to the extension's parser. Nevertheless, the reconstructed NPZ hashes differ from the archived Windows arrays, and refitting in the current Linux environment does not reproduce every original per-seed prediction. The legacy-sampler bridge reports those differences explicitly; original and new results are never pooled. Preserving the new initial fitted models, per-block results, input-array digests, and pinned environment improves the reproducibility of this extension without retroactively claiming bitwise reproduction of the October run.

7 Conclusions

Threshold adaptation is a strong alternative explanation for apparent gains from adaptive intrusion detection. In the original TON experiment, it accounts for most of the recall recovery along the frozen-score comparison, while head fitting mainly reduces FP on previously observed vectors. That aggregate benefit hides opposing FN effects across strata and fails the original joint-error criterion.

Stronger controls make the result more precise. With the full acquired-label budget available to the threshold-only detector, the spline head exchanges recall for fewer false alarms on TON in every seed. On the separate UNSW capture, the same head yields a small recall gain but substantially more false alarms, and the frozen historical spline detector remains competitive. ARF shows substantial benefit beyond its own frozen threshold control, demonstrating that the learning contribution depends on the model and its retained state. Linear and forest baselines, explicit cost-weighted operating points, and feedback sensitivity place these findings in a broader comparison. The resulting contribution is a reproducible way to identify when learning changes a useful error compromise beyond threshold adjustment, and when it does not.

Data and code availability

TON_IoT and UNSW-NB15 are supplied by UNSW under the academic-use conditions on their dataset pages [28, 29]. This study uses the network CSVs, downloaded through hash-identified public mirrors. Related TON dataset, telemetry, operating-system, architecture, and botnet publications are acknowledged as requested by the provider [30–35]. The additional UNSW-NB15 attribution papers are also cited [36–38]; their algorithms are not imported as experimental observations.

The original campaign executed repository commit `ea21691`; its retained evidence snapshot is `81d99d9` [39]. The supplied Git bundle retains the local pre-run protocol (`f4ec381`) and runner (`178e069`) history. The complete experimental evidence is published in `54871f0`, with a tree identical to the local evidence commit `33341bc`. The accompanying revision package supplies the protocol, runner, tests, pinned dependencies, input manifests, all complete per-seed reports, initial-model artifacts, analysis CSVs, figure code, and a versioned integration patch. Raw network data are not redistributed. The reproducibility README separates rebuilding the paper from

reconstructing data and rerunning experiments. A permanent public archival identifier has not been minted and must not be inferred from the Git commit identifiers.

Author contributions and declarations

CRediT statement for author review. Oleksandr Kuznetsov: Conceptualization, Methodology, Supervision, Validation, Writing—original draft, Writing—review and editing. Emanuele Pio De Bernardis: Software, Investigation, Data curation, Validation, Visualization, Writing—original draft, Writing—review and editing. The statement describes the documented division of the original project work; both authors retain responsibility for reviewing the extended experiments and final text.

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Ethics and data use. The work analyzes existing research datasets and involves no new participant recruitment or live-network attack deployment. Dataset attribution and use conditions are retained.

AI-assisted work. ChatGPT/Codex assisted with literature organization, manuscript revision, LaTeX preparation, implementing and executing the additional experiments, numerical audits, and figure generation. The experiment artifacts record the executed code and environment. The human authors must review and take responsibility for the scientific interpretation, references, authorship, and submission declarations.

A Implementation details and reproducibility map

A.1 Immutable roles and retained examples

The role hash is SHA-256 over the ASCII salt `ADAPT04-C-seed42-v1`, a zero byte, and the eight raw feature values encoded as little-endian float64. Both signs of zero are normalized to positive zero; NaNs are canonicalized. The digest is interpreted as a big-endian integer and assigned to CAL if its remainder modulo four is zero, otherwise to FIT. The salt's historical seed label is literal: the same salt is used for every final seed. Each requested observation retains its assigned role. Stored model inputs are transformed only when its label arrives; the role digest continues to identify the raw vector.

For chronological overlap, the separately defined fingerprint is SHA-256 over the 32 bytes of the eight stored float32 inputs in little-endian order. This predicate checks equality at the stored feature precision. It is not an event-identity test. The two hashing procedures have different purposes and must not be substituted for one another in reproduction.

Calibration age is measured from the observation block, not from the arrival block. Repeated vectors can occupy multiple positions in a role memory; the five-per-class minimum counts observations, not unique vectors. Class quotas determine eviction when capacity is exceeded; an age filter determines which retained CAL64 entries are eligible for threshold selection. FIT192 has no age filter. Pending requests are maintained separately from both memories.

A.2 Ordering and end-of-stream accounting

For each block, the runner (i) transforms and scores records using the incoming state, (ii) records the four decision outcomes for the evaluator, (iii) requests labels without using their values, (iv) receives the previous block's requested labels, (v) updates memories, and (vi) attempts an atomic

transaction for the next block. Labels from a predicted block do not affect any predictions within that block.

There are 355 prediction blocks but 354 update opportunities because no requested labels arrive at the end of block zero. A decision at the end of the final block is recorded and timed, although its resulting state has no subsequent evaluated block. The last 99 requested labels remain pending. This boundary explains why counts of decisions and counts of affected prediction blocks differ. Likewise, the C temporal stratification excludes the two initial prediction blocks without a preceding delayed-label decision.

A.3 Artifact entry points

The following paths are relative to `adattamento-drift/ton_temporal/` in the repository snapshot. They provide a direct map from the manuscript to implementation and evidence:

- `replay.py`: fixed transformation, additive spline model, logistic head refitting, sampling, and class-aware memory.
- `soglia_bilanciata.py`: strict-threshold convention, candidate construction, exact comparison, and lower-median tie rule.
- `calibrazione_recente.py`: immutable roles, recent-label eligibility, and atomic transaction.
- `replay_validazione_finale.py`: five-seed final replay, paired traces, overlap mask, and aggregate metric definitions.
- `split_manifest.json` and `sovrapposizioni_abcd.json`: chronology, counts, historical exposure, and cross-partition vector checks.
- `articolo_02/protocollo_validazione_finale.md`: protocol fixed before the final predictive campaign.
- `articolo_02/validazione_finale_D/analisi_finale.json`: absolute metrics, paired differences, temporal summaries, and measured costs.
- `adapt04_calibrazione_recente/replay_completo/confronto_per_blocco.csv`: C per-block evidence used in Figure 4.

The manuscript package retains these original summaries and the initial Methods/Results draft. The accompanying extension package adds the complete reports and model artifacts described in Appendix B. Rebuilding figures is separate from reconstructing raw data and rerunning the predictive campaign.

B Supplementary experimental summaries

Table 13: Extension resource ledger, mean ± seed SD. Times are seconds. Initial artifact size includes the transform; final base-model size excludes it and all external controller state. Neither size is peak RAM. Offline cached replay is not deployment latency.

Dataset	Model	Initial fit	B scoring	D cache	Replay	Initial KiB	Final base KiB
TON	ARF	179.4 ± 7.5	42.5 ± 3.2	160.0 ± 23.0	231.6 ± 8.6	938.7 ± 328.8	543.8 ± 223.9
TON	LR	0.3 ± 0.1	0.7 ± 0.1	1.3 ± 0.1	16.3 ± 1.6	1.0 ± 0.0	0.7 ± 0.0
TON	SPLINE	4.4 ± 0.4	16.4 ± 1.5	11.7 ± 0.7	25.6 ± 1.4	2.1 ± 0.0	1.9 ± 0.0
UNSW	ARF	124.6 ± 8.6	17.2 ± 1.6	29.2 ± 1.8	63.9 ± 4.5	340.5 ± 100.7	$3{,}935.9 \pm 689.1$
UNSW	LR	0.3 ± 0.0	0.1 ± 0.0	0.3 ± 0.0	4.2 ± 0.3	1.0 ± 0.0	0.7 ± 0.0
UNSW	SPLINE	3.1 ± 0.2	1.4 ± 0.1	3.0 ± 0.3	6.7 ± 0.4	2.1 ± 0.0	1.9 ± 0.0

For the resampling lengths in Table 16, the full-stream FN range crosses zero while the FP range remains negative. On unseen vectors the FP and BA ranges cross zero. This reinforces the distinction between an aggregate FP saving and stable joint improvement, but the ranges are conditional summaries of the recorded trajectories. A valid inference about retrained policies under

Table 14: Fixed-score ranking summaries in the extension, mean ± seed SD. Pooled AUROC compares all fixed scores; macro block AUROC weights valid blocks equally; normal-weighted AUROC weights them by normal observations. The full CSV includes valid-block counts and all overlap strata.

Dataset	Initial model	Pooled AUROC	Macro block	Normal-weighted block
TON	SPLINE	0.821 ± 0.020	0.850 ± 0.017	0.557 ± 0.004
TON	LR	0.318 ± 0.007	0.432 ± 0.005	0.566 ± 0.002
TON	ARF	0.425 ± 0.023	0.540 ± 0.073	0.532 ± 0.032
UNSW	SPLINE	0.973 ± 0.000	0.977 ± 0.000	0.977 ± 0.001
UNSW	LR	0.922 ± 0.066	0.921 ± 0.080	0.924 ± 0.072
UNSW	ARF	0.439 ± 0.278	0.505 ± 0.220	0.519 ± 0.215

Table 15: Attack-family recall (percent) for the spline detector in the extension. Counts are evaluated attacks, shared by every trace and seed. Means and sample SDs refer to five seeds. The machine-readable supplement includes LR and ARF, each seed, and every operating-cost policy.

Dataset	Attack family	Count	Frozen/B	Frozen/CAL256	FIT192/CAL64
TON	backdoor	508,116	99.96 ± 0.00	99.97 ± 0.00	98.78 ± 1.16
TON	mitm	1,052	4.81 ± 0.33	12.79 ± 1.93	4.77 ± 1.60
TON	password	549,383	19.81 ± 0.01	86.08 ± 3.38	74.39 ± 4.04
TON	ransomware	72,805	9.29 ± 0.00	72.33 ± 14.59	54.56 ± 2.97
TON	xss	2,108,944	2.60 ± 0.57	92.92 ± 1.49	92.38 ± 1.06
UNSW	Analysis	1,459	93.83 ± 0.44	94.37 ± 0.46	93.78 ± 1.24
UNSW	Backdoor	1,390	86.37 ± 0.38	90.04 ± 0.29	92.75 ± 0.41
UNSW	DoS	10,262	86.91 ± 0.39	91.23 ± 0.23	93.83 ± 0.35
UNSW	Exploits	26,423	82.72 ± 0.48	89.05 ± 0.31	90.12 ± 0.61
UNSW	Fuzzers	13,334	76.22 ± 0.47	82.85 ± 0.57	85.01 ± 1.11
UNSW	Generic	149,998	99.45 ± 0.02	99.63 ± 0.01	99.65 ± 0.02
UNSW	Reconnaissance	8,426	87.56 ± 1.51	92.54 ± 0.78	92.36 ± 0.94
UNSW	Shellcode	886	60.93 ± 0.57	77.09 ± 1.15	77.22 ± 1.56
UNSW	Worms	104	72.88 ± 1.43	85.00 ± 2.11	83.65 ± 4.30

Table 16: Descriptive moving-block sensitivity of the original paired FIT192/CAL64 minus Frozen/CAL64 errors. Brackets are 2.5th–97.5th resampling percentiles, not confidence intervals for a refitted adaptive process. Each analysis resamples contiguous blocks of the recorded error trajectories, sharing block selections across seeds and policies.

Scope	Block length	ΔFN (thousands)	ΔFP (thousands)	ΔBA (pp)
All D	5	[-52.96, 35.87]	[-138.85, -32.64]	[8.77, 16.88]
All D	20	[-73.68, 58.34]	[-190.30, -9.77]	[4.97, 17.72]
All D	50	[-94.44, 85.35]	[-217.51, -8.33]	[5.19, 16.47]
Unseen	5	[-94.51, -47.32]	[-19.38, 13.64]	[-3.49, 8.42]
Unseen	20	[-104.59, -36.07]	[-19.62, 19.53]	[-4.19, 10.15]
Unseen	50	[-116.03, -19.18]	[-18.79, 17.03]	[-3.07, 11.29]

a new process would require a sampling design for that process and rerunning the adaptive state transitions.

B.1 Numerical bridge and archive contents

The legacy-sampler reruns use exactly the original requested row identities in every block. Their mean FIT192/CAL64 outcomes are close to the original aggregates (the FN difference is −563.0 and the FP difference is −7.4), but individual seeds and some other traces differ materially. For example, the FIT192/B FN difference ranges from −9,617 to +335,381. Accordingly, the original per-seed observations and the newly fitted observations are separate evidence stages. The file `analysis/legacy_replication_bridge.csv` gives every difference; no rerun is presented as bitwise replication.

The original A parser and the extension A parser produce identical arrays and an identical NPZ file in the present environment. The extension also records array-level digests to distinguish data identity from archive-container bytes. Repeating the initial seed-42 fit with NumPy 2.3.4 and SciPy 1.16.3 in Linux gives the same fitted head as the new environment, so library version labels alone do not identify the source of the Windows/Linux difference. An exact causal attribution would require the original arrays and fitted model state. This residual reproducibility boundary is retained rather than explained speculatively.

The revision archive includes all 55 extension reports, the original five full reports, compact fitted initial-model artifacts, input manifests, and a validation summary. `analysis/per_seed_metrics.csv` contains every controller, seed, condition, and overlap stratum; `analysis/paired_differences.csv` preserves paired comparisons. `analysis/attack_family_recall.csv`, `analysis/resources_and_labels.csv`, and `analysis/time_partitions.json` document attack counts, computational components, feedback timing, and chronology. All paper figures and generated tables can be rebuilt from these artifacts without downloading the raw CSVs.

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